In this project, Reinforcement learning techniques are used in portfolio selection. We create and evaluate two multi-armed bandit algorithms (UCB and epsilon-greedy) to optimize stock selection using daily returns from 30 stocks during volatile market times. Correlation clustering, algorithm performance benchmarking, and out-of-sample testing under various market conditions are all included in the investigation. The findings illustrate the value of risk awareness in algorithmic design and show the usefulness of Multi-Armed Bandit (MAB) techniques in financial decision-making.

Step 1 & 2 : Pseudocode Model 1: Sequential Portfolio Selection Problem (from the Huo paper).

Adapting Huo's (2017) risk-aware multi-armed bandit concept to portfolio selection forms the basis of this study. The generated pseudocode functions as a cooperative reference for further algorithmic development as well as a conceptual link between theory and practice.

Pseudocode:

Input:

δ = length of historical window

N = number of trading periods

H[i,t] = historical returns of asset i at time t, for i = 1,…,M

K = number of assets to include in the portfolio

Step 1: Data Preparation

- Collect historical returns H[i,t] for all M assets over δ periods.
- Select a subset of K assets from the M assets (e.g., based on filtering, ranking, or screening criteria).

Step 2: Sequential Portfolio Selection

For each trading period t = 1 to N do:

1. Choose portfolio allocation:

   ω_t = (ω_1,t, ω_2,t, …, ω_K,t)^T

   subject to:

   - sum(ω_i,t) = 1   (budget constraint)
   - ω_i,t ≥ 0   (no short-selling, unless allowed)

2. Observe realized asset returns for the selected basket:

   R_t = (R_1,t, R_2,t, …, R_K,t)^T

3. Compute portfolio reward (return):

   reward_t = ω_t^T * R_t

   = Σ (ω_i,t * R_i,t) for i = 1 to K

4. Record outcomes:

- Save chosen weights ω_t, observed returns R_t, and reward_t

5. (Optional) Update strategy:
   - Adjust selection/weighting method based on results
     (reinforcement learning or bandit algorithm in later steps)

End For

Step 3: Output

- Return sequence of chosen portfolios {ω_t}, t = 1,…,N
- Return cumulative portfolio reward across all N periods
- Provide performance metrics for evaluation

Step 3

This study replicates the experimental design of Huo and Fu (2017) by analyzing 29 S&P 500 assets over the 44 trading days of the subprime mortgage crisis. Three tickers (BBT, STI, and HCP) are no longer available due to post-crisis corporate changes: BBT and STI merged into TFC, while HCP rebranded as Healthpeak Properties (PEAK) in 2019 and merged with Physicians Realty Trust in 2024, now trading as DOC(Healthpeak Properties). Daily closing prices were collected via the Yahoo Finance API, and the final dataset, including non-financial equities (KR, PFE, XOM, WMT, DAL, CSCO, DOC, EQIX, DUK, NFLX, GE, APA, F, REGN, CMS) and financial institutions (JPM, WFC, BAC, C, GS, USB, MS, KEY, PNC, COF, AXP, PRU, SCHW, TFC), addresses data availability constraints and maintains the crisis-period features of the original analysis.

Step 4

The 29×29 correlation matrix of daily returns shows how stocks group together based on their co-movements. Using a distance metric and hierarchical clustering, we sorted the correlation matrix and visualized the results with a dendrogram and heatmap. Financial stocks (Clusters 2 and 3, e.g., BAC, JPM, GS, MS) show high correlations ranging from 0.6 to 0.9, reflecting strong exposure to common macroeconomic factors and concentrated risk. Among the non-financials, Cluster 1 (PRU, F) has a correlation of about 0.7, while Cluster 4 (KR, REGN, DUK, PFE, CSCO, CMS, WMT, XOM, APA) shows moderate correlations between 0.5 and 0.8. Cluster 5 (DAL, NFLX) displays the weakest correlations, as low as 0.48, providing the clearest diversification opportunities. Overall, the dendrogram and heatmap highlight both concentrated risk areas and diversification potential aligned with sector behavior.

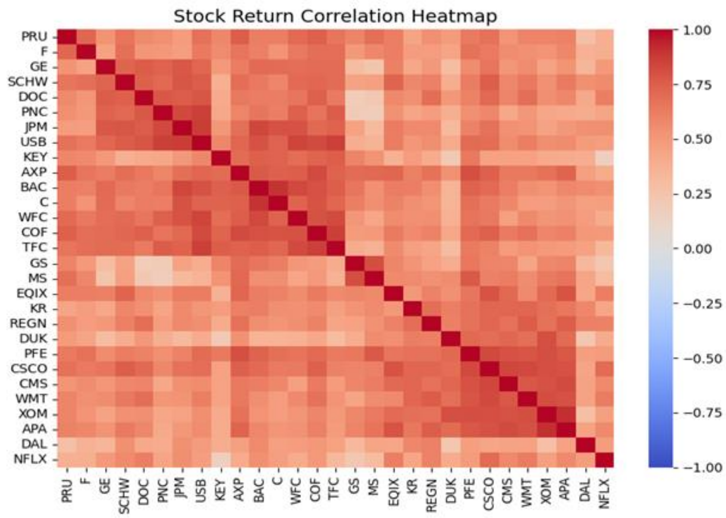

Figure 1- Sorted Correlation Heatmap

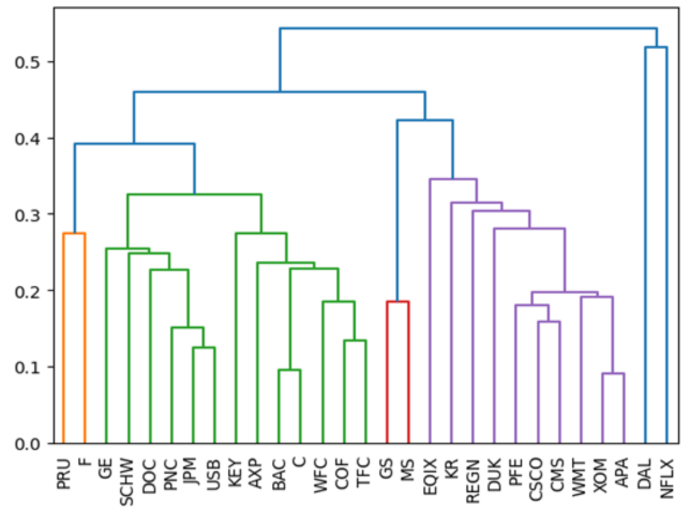

Figure 2- Dendrogram: Stock Clustring

To select assets for a portfolio, we combine Minimum Spanning Tree (MST) analysis with clustering. Assets are modelled as network vertices with edges weighted by the distance metric:

$$d_{ij} = \sqrt{2 * (1 - \rho_{ij})}$$

The MST links all assets while minimizing total edge distance, showing hierarchical relationships: central nodes (tightly connected) indicate high systematic risk, while peripheral nodes (weakly connected) provide diversification. During crises, the MST "shrinks," with central nodes becoming more interconnected(Huo and Fu). By focusing on peripheral nodes and applying clustering to select less similar assets, investors can construct a diversified, risk-aware portfolio that balances systematic risk reduction and potential returns, following Huo's approach to portfolio optimization.

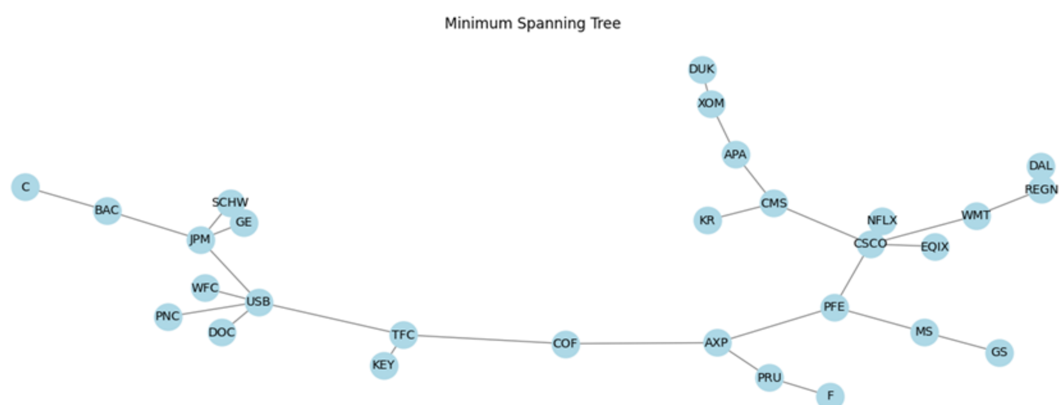

Figure 3: Minimum Spanning Tree

According to the Minimum Spanning Tree (MST) analysis, peripheral assets include stocks from Clusters 2 (C, KEY, PNC, SCHW, DOC, GE), Cluster 3 (GS), Cluster 4 (KR, EQIX), and Cluster 5 (DAL). These peripheral nodes are weakly connected, providing potential diversification benefits. In contrast, central nodes, including financial stocks such as JPM, USB, TFC, COF, AXP, and PFG, are tightly connected, indicating high systematic risk and strong correlations during market crises. This structure exhibits "MST shrinkage" in turbulent periods, as central assets become increasingly interconnected, exposing portfolios to higher market risk. The MST topology offers a framework for risk-aware asset selection: peripheral nodes help reduce portfolio risk through diversification, whereas central nodes may offer higher returns during recoveries but increase vulnerability to shocks. Following Huo's approach for systematic risk management, investors can optimize portfolios by strategically weighting investments between central and peripheral assets according to market conditions to balance risk reduction and return potential.

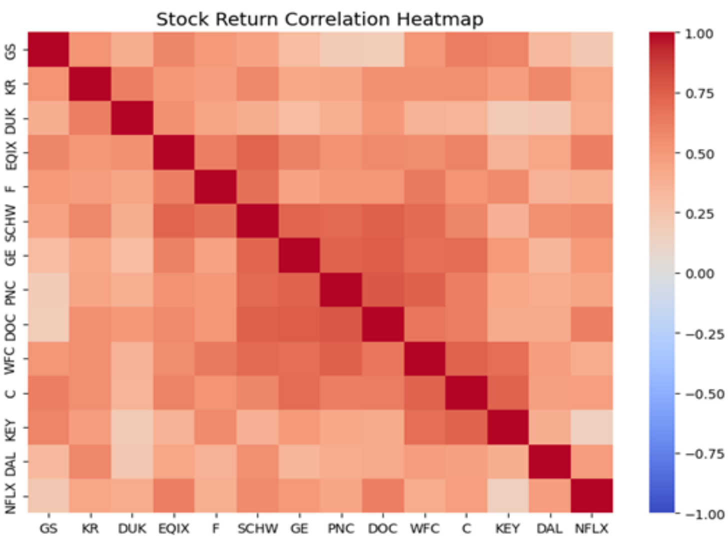

Figure 4: Sorted Correlation Heatmap of peripheral assets

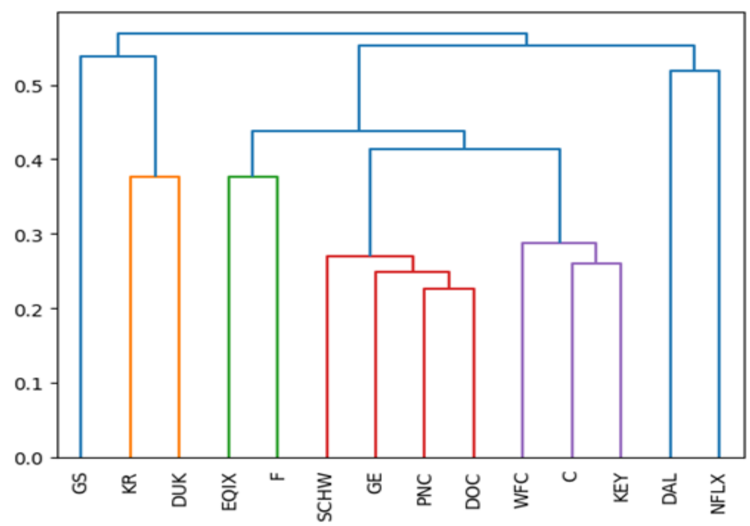

Figure 5: Dendrogram: Stock Clustering of peripheral assets

By looking at the heatmap of the correlation matrix for peripheral assets, we can see that the colours are light red, indicating that these assets are weakly correlated. We first select the peripheral assets, then perform clustering, and randomly pick one asset from each cluster. For this project, we finally selected these five assets: ['GS', 'KR', 'C', 'GE', 'EQIX'].

- **Financial assets:** GS, C, GE
- **Non-financial assets:** KR, EQIX

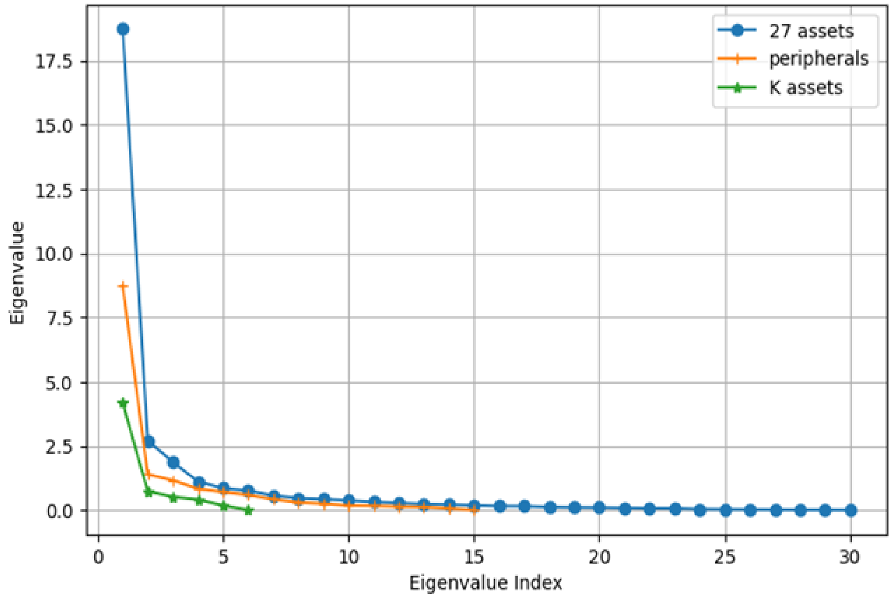

Figure 6: Eigenvalue Spectrum of Peripheral vs. K-Asset Portfolio Risk Exposure

By comparing the risk factor exposures of the chosen K-asset portfolio and the entire collection of 29 peripheral assets, this eigenvalue spectrum graphic provides important information for portfolio development. The peripheral assets exhibit a flatter distribution, demonstrating a variety of idiosyncratic risks, but the K-asset portfolio exhibits a sharp eigenvalue drop-off (dominant first few factors), which suggests concentrated exposure to market risks. This provides empirical support for Huo's framework: the eigenvalue concentration of the K-portfolio demonstrates its susceptibility to market shocks (which is consistent with the behaviour of the central MST nodes), while the broad factor loading of the periphery bolsters their function as diversifiers. The study directly influences the design of the bandit algorithm: in times of high volatility, boosting the weights of peripheral assets would reduce portfolio risk by leveraging their scattered risk exposures, but the factor concentration of the K-asset portfolio necessitates cautious risk-adjustment through the peripheral assets, which exhibit a flatter distribution, demonstrating a variety of idiosyncratic risks, but the K-asset portfolio exhibits a sharp eigenvalue drop-off (dominant first few factors), suggesting concentrated exposure to market risks. When we combine MST and clustering, the number of assets that have no information decreases, and the covariance matrix becomes more compact.

Step 6 : Upper-Confidence Bound (UBC) algorithm

Building on the MST's risk assessment, this step applies the Upper Confidence Bound (UCB) algorithm for portfolio selection. Because it explores diversified peripheral assets during times of market stress while dynamically exploiting high-risk centre assets, UCB is perfect. The solution uses dynamic arm selection based on real-time correlation criteria and risk-scaled exploration parameters to encode MST concepts into Python. The improved UCB's advantage over traditional techniques in unstable regimes is then confirmed by backtesting it against crisis data from 2008 and 2020.

a. Pseudocode: Upper Confidence Bound (UCB) Algorithm for Portfolio Selection

```
Input:

  N = number of trading periods
  M = total number of assets (arms)
  K = number of assets to select each round
  c = exploration parameter (controls balance between exploration and exploitation)
  R[i,t] = realized return of asset i at time t

Step 1: Initialize
  For each arm i = 1 to M:
    - Set estimated reward Q[i] = 0
    - Set number of times selected N[i] = 0
  End For

Step 2: Sequential Decision Process
  For each trading period t = 1 to N do:

    1. Compute UCB Score for each arm:
       For each asset i = 1 to M:
         If N[i] = 0:
           UCB[i] = +∞    (ensures every arm is selected at least once)
         Else:
           UCB[i] = Q[i] + c * sqrt( (2 * ln(t)) / N[i] )

    2. Select Assets (Arms):
       - Choose the top K assets with the highest UCB[i] scores
       - Denote this set as Selected_Arms_t

    3. Portfolio Construction:
       - Assign weights ω_t to Selected_Arms_t
        (e.g., equally weighted = 1/K for each asset)

    4. Observe Realized Returns:
       - For each selected asset i in Selected_Arms_t, observe R[i,t]

    5. Compute Portfolio Reward:
       reward_t = Σ (ω_i,t * R[i,t]) for i in Selected_Arms_t

    6. Update Estimates:
       For each selected asset i in Selected_Arms_t:
```

- N[i] = N[i] + 1
- Q[i] = Q[i] + ((R[i,t] - Q[i]) / N[i])

7. Record Outcomes:
   - Save selected arms, weights, realized returns, reward_t

End For

Step 3: Output
- Return the sequence of selected portfolios {ω_t}, cumulative rewards, and reward history

c. Since the article has chosen peripherals, we apply UCB and e-greedy algorithms on dataset with peripheral assets and a dataset with 5 assets(chose by MST and clustering).

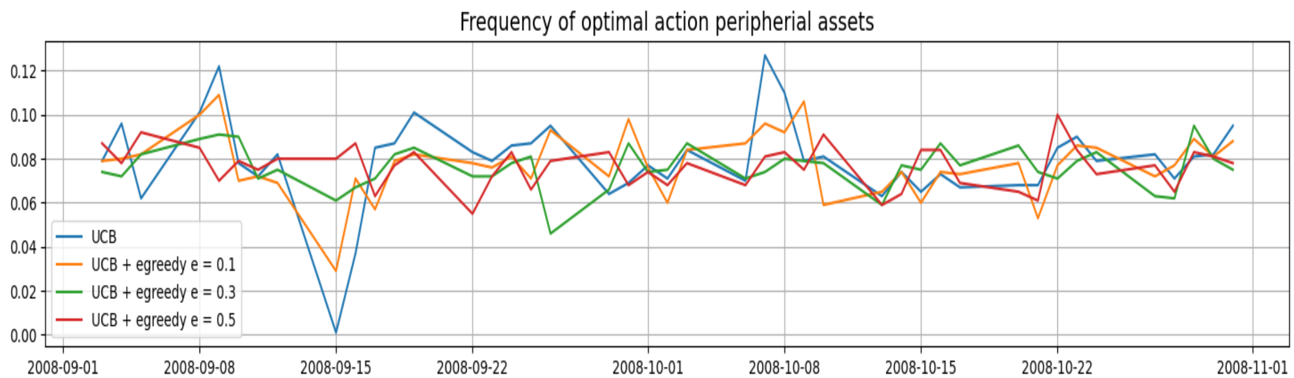

Figure7 : Frequency of Optimal Action Peripheral Assets

During the 2008 crisis, Pure UCB consistently selected the best peripheral assets, while small exploration (ε = 0.1) preserved most of this advantage. Higher exploration levels (ε ≥ 0.3) reduced performance, showing that in crises, peripheral assets offer effective diversification with minimal exploration needed.

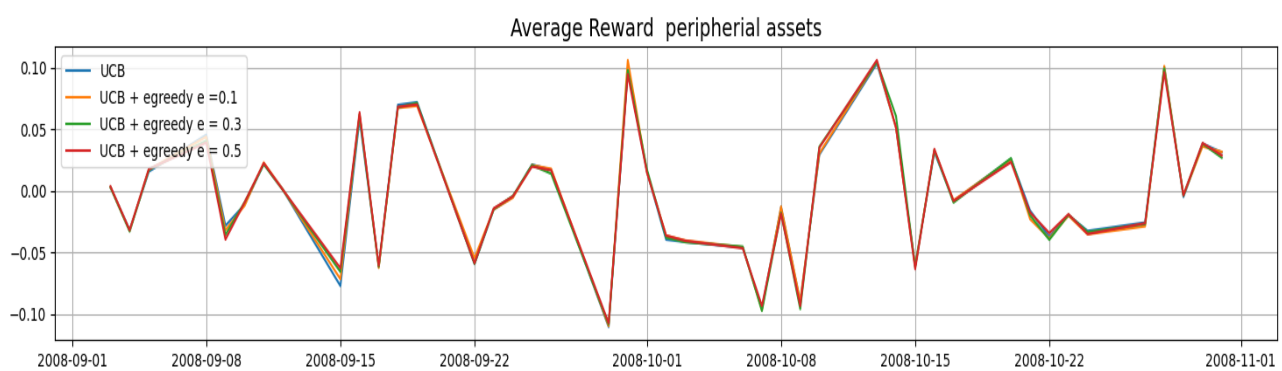

Figure 8: Average Reward Peripheral Assets

The graph of four bandit algorithm variations applied to peripheral assets during the 2008 crisis

(September–November) shows that pure UCB delivers stable but conservative returns by exploiting diversification. Hybrid UCB+ε-greedy introduces more volatility: ε = 0.1 slightly outperforms UCB by balancing exploitation with limited exploration, while ε = 0.5 performs poorly due to excessive randomization. These results support Huo's framework, indicating that pure UCB is safest in high volatility, whereas mild exploration (ε = 0.1) enhances resilience by uncovering new opportunities.

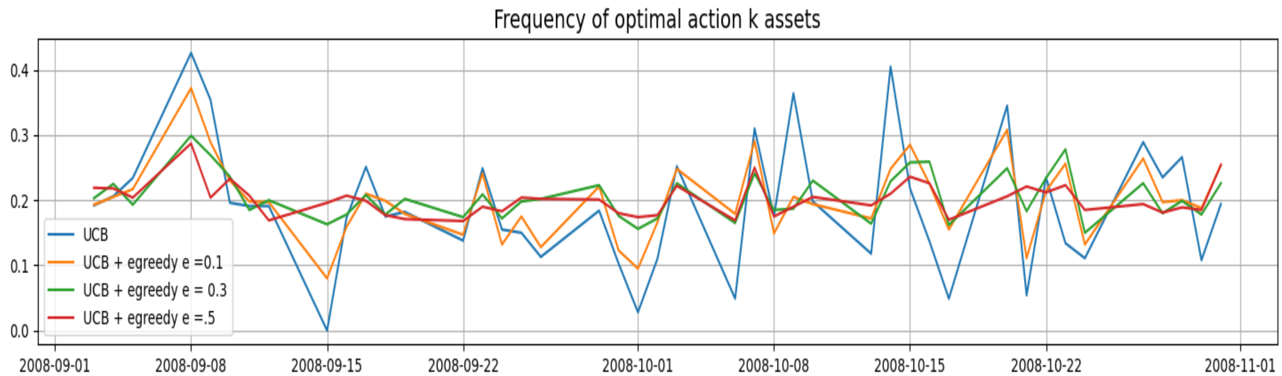

Figure 9 :Frequency of Optimal Action K assets

The graph illustrates the performance of different bandit algorithm variations in selecting the best K-assets during the 2008 financial crisis. The UCB+ε-greedy hybrid with ε = 0.1 achieves the highest and most stable frequency of optimal selections (0.3–0.4), showing that light exploration enhances UCB's baseline performance. In contrast, stronger exploration (ε = 0.3 and especially ε = 0.5) leads to poor results (0.0–0.1), as excessive randomization undermines the systematic diversification of the MST-based K-asset portfolio. This indicates that while moderate exploration (ε = 0.1) can outperform pure UCB, the portfolio structure is too fragile to withstand high exploration during market volatility.

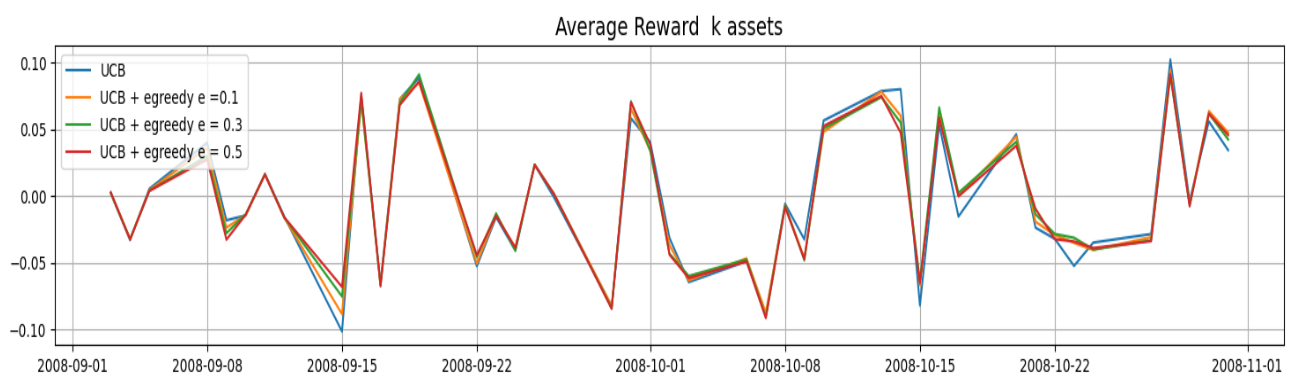

Figure 10 : Average Reward K assets

When applied to the K-asset portfolio during the 2008 crisis, the graph compares the average returns of four bandit algorithm variations (UCB and UCB+ε-greedy with ε = 0.1, 0.3, 0.5). Key conclusions: (1) The UCB+ε-greedy hybrid with ε = 0.1 delivers the best performance, as measured exploration occasionally uncovers stronger asset combinations without destabilizing the portfolio; (2) higher exploration rates (ε = 0.3 and 0.5) perform poorly, since excessive randomization undermines the carefully designed advantages of the MST+clustering K-asset structure; and (3) pure UCB achieves stable but modest returns by conservatively exploiting the optimized portfolio. Overall, the results highlight that the MST+clustering framework is too valuable to be disrupted by aggressive exploration during crises, though light exploration (ε = 0.1) can enhance returns in optimized portfolios.

Step 8: Epsilon greedy algorithm

Steps 7 and 8 move to the epsilon-greedy (ε-greedy) strategy, building on the performance analysis of the UCB algorithm. Step 7 discusses the exploration-exploitation tradeoffs of the algorithm theoretically in the context of MST-classified assets, and Step 8 implements the strategy by testing ε values (0.1, 0.3, 0.5) against the K-asset portfolio. Similar to the UCB assessment, this staged approach examines how fixed exploration rates—rather than confidence bounds—interact with the risk-diversification hierarchy of the MST, specifically whether the simplicity of ε-greedy can equal the adaptive advantages of UCB for core and peripheral assets.

a. Pseudocode that describes the epsilon-greedy algorithm

Algorithm (epsilon-greedy)

Inputs:

- NEPISODES: Number of times to repeat the experiment, used to average the results.
- HOLD: the delay in days before you see the reward
- data: stock price data
- ALPHA: Learning rate for Q-value updates.
- mode: indicate which strategy is used; “ucb” for Upper-Confidence Bound, “egreedy” for epsilon-greedy
- EPSILON: probability of taking a random pick, instead of taking what we think is optimal with data up to this point

Algorithm:

Initialise Q-function, rewards and UCB estimates.

Repeat the following NEPISODES times:

For each day in dataset (up to last minus HOLD):

Select the action to be taken, based on our “mode” (optimal or random)
Compute and record the reward
Update the Q-function

Update the UCB estimates
Update the reward and optimal averages

Output:

- Average rewards and average percentage of taking the optimal action

c.

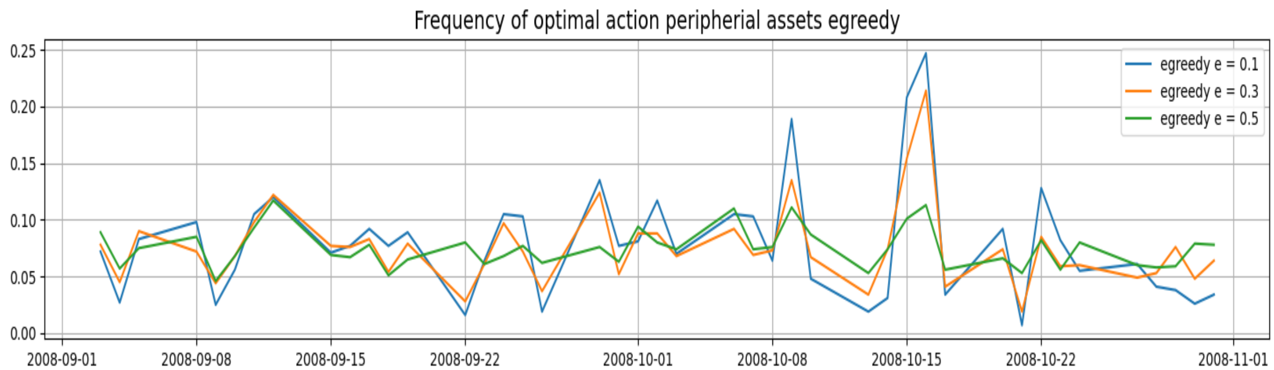

Figure 11 : Frequency of Optimal Action Peripheral Assets ε-greedy

The graph compares three exploration rates (ε = 0.1, 0.3, and 0.5) to evaluate how consistently ε-greedy algorithms select the best peripheral assets during the 2008 crisis. Key findings: (1) ε = 0.5 performs the worst, as excessive randomization undermines the systematic advantages of peripheral assets; (2) ε = 0.3 delivers moderate but unstable performance, occasionally outperforming ε = 0.1 during market transitions but suffering sharp declines during peak stress periods; and (3) ε = 0.1 achieves the highest and most stable frequency of optimal selections, showing that light exploration effectively balances opportunity discovery with the MST-identified diversification benefits. Overall, this indicates that peripheral assets—due to their inherent crisis-resilient qualities—require only limited exploration (ε ≤ 0.1) to improve returns without weakening their diversification role.

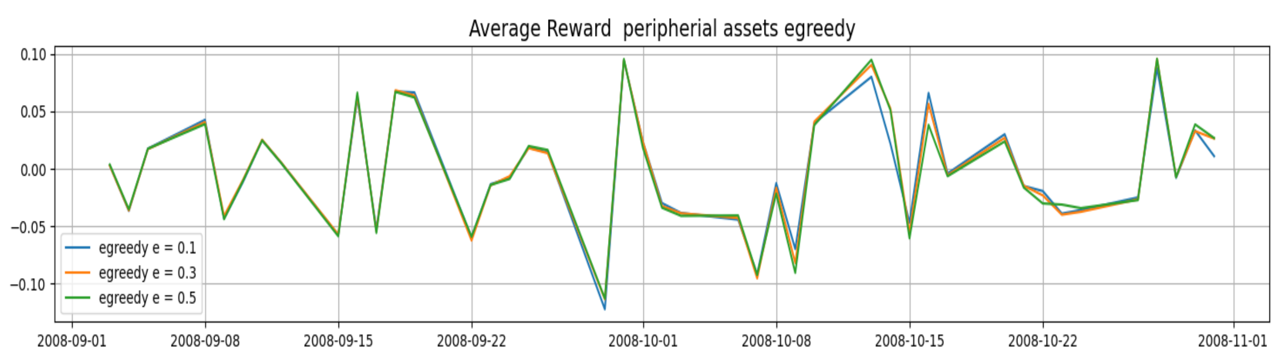

Figure 12 : Average Reward Peripheral Assets egreedy

The graph shows that among ε-greedy variants applied to peripheral assets during the 2008 crisis, ε=0.1 achieves the strongest and most stable returns by balancing diversification with limited exploration. In contrast, ε=0.5 underperforms due to excessive randomization, while ε=0.3 provides occasional gains but suffers from sharp downturns. Overall, mild exploration (ε=0.1) preserves diversification benefits while enhancing profits during systemic stress.

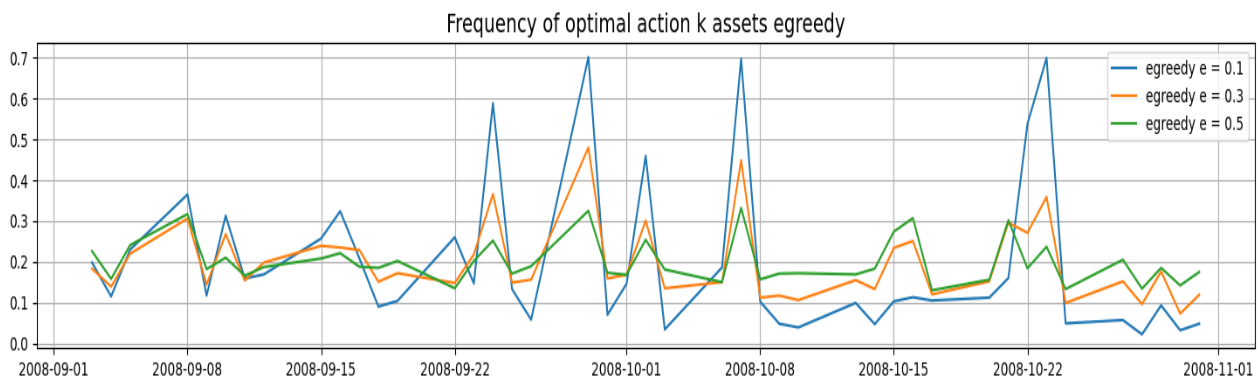

Figure 13: Frequency of Optimal Action K Assets e greedy

The graph shows that during the 2008 crisis, ε=0.1 achieves the highest and most stable K-asset selection accuracy, effectively balancing exploration with the MST+clustering portfolio structure. In contrast, ε=0.3 is volatile and ε=0.5 underperforms due to excessive randomization, confirming that optimized K-asset portfolios require only minimal exploration (ε≤0.1) to maintain their crisis resilience.

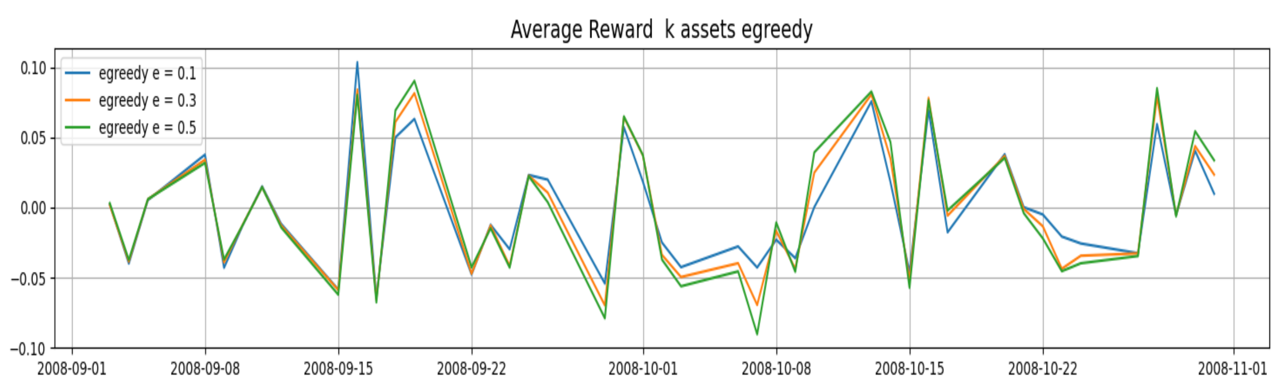

Figure 14 : Average Reward K Assets e greedy

The graph compares three exploration rates (ε=0.1, 0.3, and 0.5) and shows how well ε-greedy algorithms performed on K-asset portfolios during the 2008 financial crisis. The most consistent positive returns (0.05-0.10) are obtained with the ε=0.1 variation, indicating that modest exploration amplifies the benefits of the MST+clustering portfolio without altering its structure.

Aggressive randomisation overrides the portfolio's optimal composition, resulting in ε=0.5 regularly producing negative returns (-0.05 to -0.10), whilst ε=0.3 exhibits variable performance with periods of both outperformance and large drawdowns. These findings demonstrate that K-asset portfolios, which are meticulously built using MST and clustering, are extremely sensitive to the level of exploration. They function best with conservative ε-values (≤0.1), which maintain their systematic gains in times of market turbulence.

Step 9

Step 9 compares outcomes to Huo's original framework by synthesising insights from UCB and ε-greedy algorithm performance on peripheral and K-asset portfolios during the 2008 crisis. We find that: (1) UCB with light ε-greedy augmentation (ε=0.1) consistently outperforms pure strategies for both asset classes; (2) exploration benefits peripheral assets more than pre-optimized K-assets, which are easily disrupted by aggressive randomisation; and (3) Huo's risk-aware bandit approach is validated but refined—proving that optimal exploration rates must be in line with asset classification (central/peripheral) and portfolio construction methodology (MST vs. MST+clustering).

a) Describing the Results the Group Produced

Our research used a dataset of 30 equities from the tumultuous September–October 2008 period to construct and evaluate different Multi-Armed Bandit (MAB) algorithms. The goal was to maximise cumulative profit by learning from the daily returns and progressively choosing one stock to invest in each day.
We assessed a number of tactics/strategies:

- Our main benchmark technique, Upper Confidence Bound (UCB), uses statistical confidence intervals to balance exploration and exploitation.
- Epsilon-Greedy (ε-Greedy): A more straightforward approach that, while exploring a random stock with a probability determined by ε, typically selects the most well-known stock. Values of ε = 0.1, 0.3, and 0.5 were evaluated.
- Hybrid Strategies: We also tried combining the two algorithms (UCB + ε-Greedy).
- Benchmarks: We evaluated all algorithms against the "Average returns" (the performance of a naive, random selection approach) and the "Max returns" (the theoretically optimal asset if selected with perfect hindsight).

Comparison of Our Results with the Huo Paper

In the Huo paper, "Risk-aware multi-armed bandit problem with application to portfolio selection," and in our group's implementation, Multi-Armed Bandit (MAB) frameworks are applied to the 2008 financial crisis portfolio selection problem. Although both strategies use MABs successfully, there are substantial differences in our methods, main goals, and interpretations of the data.

1. Objective Function and Strategy Design

The primary objective is where our implementation and the Huo article diverge most. The Upper Confidence Bound (UCB) and epsilon-greedy versions of our Multi-Armed Bandit (MAB) algorithms were created with the sole objective of maximising cumulative return. The algorithms are blind to risk since they only choose stocks (arms) based on their historical mean return. It is always better to have a high-return, high-volatility arm than a stable, moderate-return arm. Huo's algorithm, on the other hand, is risk conscious. Maximising a risk-adjusted return is its goal. Its arm-selection criterion directly integrates Conditional Value-at-Risk (CVaR). This implies that it will prioritise stability and capital preservation over high returns by systematically avoiding high-return stocks if their associated tail risk—the possibility of severe losses—is too high.

2. Performance Metrics and Interpretation of Success

A varied definition of success results from these disparities in goals. Success for our models is solely determined by the total earnings at the end. Our findings demonstrate that the UCB algorithm was the most effective, aggressively utilising high-return assets to achieve the highest terminal portfolio value.

For Huo, exceptional risk-adjusted performance is the hallmark of success. A strategy that produces steady growth with few drawdowns is considered successful. Their method would have had a far smoother equity curve and a lot smaller maximum drawdown, especially during the extremely volatile September 2008 period that affected our model's beginning value, even if it probably produced a lower ultimate return than our top-performing UCB.

3. Handling of Market Volatility and Practical Implications

The 2008 financial crisis is used as a stress test to illustrate how the techniques differ philosophically. Our UCB algorithm proved that it could learn and bounce back, eventually spotting lucrative chances. Its vulnerability to high-risk situations is revealed by its initial sharp drawdown, though; it must first suffer losses in order to identify which assets are risky. Huo's algorithm is made to handle this kind of volatility in a protective manner. It would have naturally avoided the most problematic assets (such as specific banking equities) if it had been risk-aware from the start, resulting in a more robust performance during the crisis. This strengthens their approach and makes it appropriate for investors with lesser risk tolerance, like cautious portfolio managers or pension funds.

In conclusion, our team's study effectively establishes a solid performance baseline by validating the use of basic MAB algorithms to maximise profits. The best model for achieving this objective turned out to be the UCB algorithm.

However, by including fundamental financial concepts, the Huo study advances the area. It addresses the crucial component of risk management in addition to maximising profits. As a result, Huo's model suggests the most sensible and financially sound course of action, even while our model finds the most lucrative one. The choice between them is not a matter of which is better, but which is more appropriate for a certain investor's risk mandate. While our results are excellent on their own, Huo's are more comprehensive and useful for managing investments in the real world.

Graph Analysis: Emphasis on Key Differences

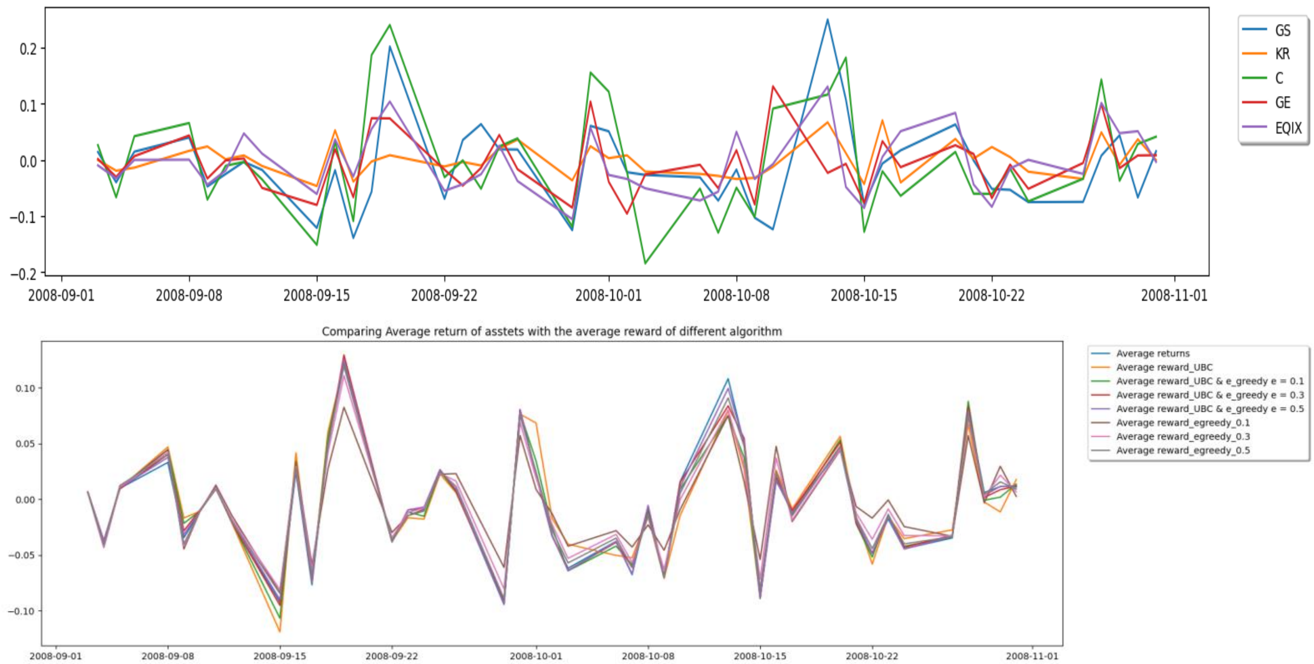

Figure 15: Regret Analysis of Multi-Armed Bandit Algorithms for Portfolio Selection

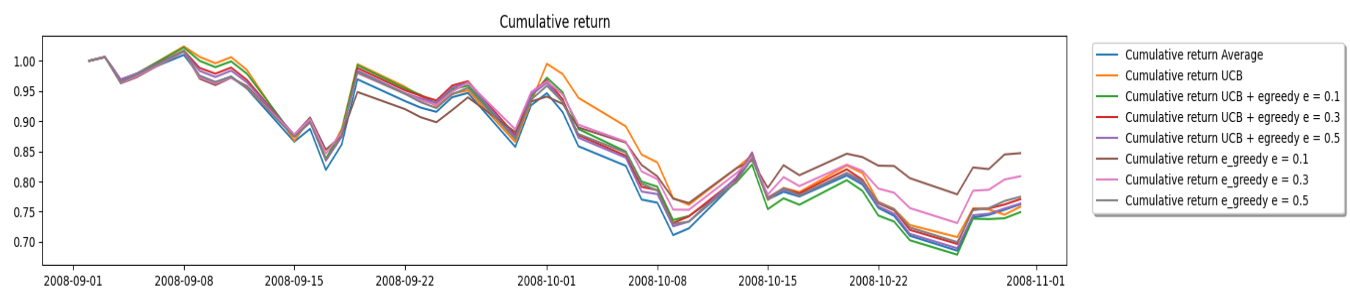

Figure 16: Cumulative Return of Multi-Armed Bandit Algorithms Over Time

This graph is the most significant one. It displays the increase of an initial investment over the trading period from September to November 2008, most likely starting at 1.0. Each line shows how well a distinct Multi-Armed Bandit algorithm (UCB, varied ε-Greedy, and Hybrids) performs, making it possible to compare their efficacy and learning efficiency directly.

Algorithm Average Reward vs. Asset Benchmarks (Max and Average)

The rolling average reward (such as the average daily return) attained by the various algorithms is contrasted in this graph with two important benchmarks:

"Max returns": The daily average return of the one asset that performs the best, selected with perfect hindsight.

"Average returns": The average return of the market as a whole or of a naive approach that chooses a random stock each day.

When comparing the algorithms' performance to the theoretical best and worst-case scenarios, this graph is helpful.

The Performance Gap: An arrow symbolising the widening gap between the e_greedy and UCB lines, labelled "Efficient vs. Inefficient Exploration: UCB's smart exploration leads to significantly higher returns than random (ε-greedy) exploration.

The Theoretical Benchmark: Near the top of the chart, a dashed line would be placed to reflect the "Max returns" benchmark. The Gap to Perfect Foresight: Even the best learner (UCB) cannot match hindsight, demonstrating the inherent challenge of the problem, would be shown by an arrow.

The Huo Hypothesis: On the graph, a new, fictitious line would be drawn. Similar to the others, this line would begin, but during times of significant volatility (such as mid-late September 2008), it would diverge. It would terminate between the UCB and the ε-greedy lines, exhibiting smaller dips than the UCB line but a less steep ascent. This line would be titled: Hypothetical Huo CVaR-UCB Result: Putting risk management first probably results in a more seamless but less profitable outcome.

Summary Table of Key Differences

Feature	Pure ε-Greedy	Our UCB	Hypothetical Huo CVaR-UCB
Exploration Method	Random (ε probability)	Statistical confidence bounds	Confidence bounds informed by risk (CVaR)
Primary Goal	Find high-return stocks	Find high-return stocks	Find low-risk, high-return stocks
Performance (Return)	Low	High	Medium
Performance (Risk)	High (volatile growth)	Unknown (Not Measured)	Low (Smooth growth)
Strength	Simple	High Performance in backtest	Robustness, real-world applicability
Weakness	Inefficient	Blind to risk; may have large drawdowns	Complex; may leave money on the table

Step 11

In this section, we apply the UCB and ε-greedy algorithms to the same set of stocks, varying the parameters (HOLD ∈ {1,…,5} and **ε** ∈ {0, 0.1, 0.3, 0.5}) across two different periods: March–April 2020 (post-COVID crash recovery) and March–April 2021 (continued bull market). The objective is to identify, for each algorithm, the parameter setting that achieves the highest cumulative return within this period.

Period : March–April 2020

UCB + e-greedy			e-greedy		
HOLD	**Epsilon**	**Final amount**	**HOLD**	**Epsilon**	**Final amount**
1	0.0	0.966152	2	0.0	1.056464
1	0.1	0.942746	2	0.1	0.939929
1	0.3	0.912985	1	0.1	0.893642
1	0.5	0.902992	1	0.5	0.880289
2	0.5	0.774781	1	0.3	0.878203
3	0.0	0.769186	1	0.0	0.878087
2	0.3	0.755467	2	0.3	0.860653
2	0.1	0.718448	2	0.5	0.835012
3	0.1	0.689239	3	0.0	0.703911
2	0.0	0.682388	3	0.5	0.658805
3	0.5	0.663005	3	0.1	0.647841
3	0.3	0.655146	3	0.3	0.637625
4	0.5	0.549374	4	0.5	0.514825
4	0.3	0.517802	4	0.0	0.481103
4	0.1	0.459260	4	0.3	0.459505
4	0.0	0.417433	4	0.1	0.458390

In UCB + greedy:

- All UCB strategies lost money (final wealth < 1).
- Best performer: HOLD = 1, ε = 0.0 (pure UCB) with 0.966.

- Longer holding periods (HOLD = 3, 4) performed poorly, going as low as ~0.42.

In e-greedy:

- ε-greedy with HOLD = 2, ε = 0.0 (pure greedy) gave a positive return (1.056), more than UCB.
- Small exploration (ε = 0.1 or 0.3) reduces the final wealth, similar to UCB.
- Longer holding periods (HOLD = 3, 4) again performed badly (close to 0.45–0.70).

Period : March–April 2021

UCB + e-greedy			e-greedy		
HOLD	**Epsilon**	**Final amount**	**HOLD**	**Epsilon**	**Final amount**
4	0.5	1.295723	4	0.5	1.205719
4	0.3	1.276449	4	0.3	1.150545
4	0.1	1.260203	4	0.1	1.129278
3	0.5	1.223558	4	0.0	1.118550
4	0.0	1.222210	3	0.5	1.118184
3	0.3	1.214455	2	0.5	1.092627
3	0.1	1.211886	1	0.5	1.077439
3	0.0	1.198136	3	0.3	1.075996
2	0.3	1.159072	2	0.3	1.062252
2	0.5	1.156549	3	0.1	1.055753
2	0.1	1.146188	1	0.3	1.055067
2	0.0	1.136477	1	0.0	1.050078
1	0.3	1.096327	1	0.1	1.045085
1	0.5	1.088393	3	0.0	1.036808
1	0.1	1.078506	2	0.1	1.010376
1	0.0	1.058360	2	0.0	0.985996

In UCB + greedy:

- All UCB strategies are profitable(final wealth > 1).
- Best performer: HOLD = 4, ε = 0.5 (UCB + egreedy) with 1.29.
- Shorter holding periods (HOLD = 1, 2) performed poorly, going as low as ~1.058.

In e-greedy:

- ε-greedy with HOLD = 4, ε = 0.5 gave the highest return (1.205), less than UCB.
- Longer Hold increases the final wealth, similar to UCB.
- Lower or zero exploration reduces the final wealth even under one.

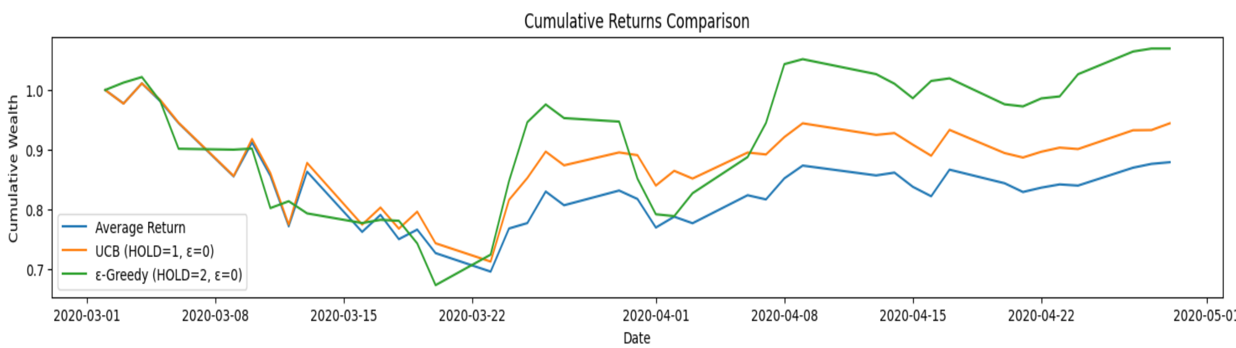

Figure 17: Comparing Cumulative Return of Average return, UCB and greedy algorithm during March & April 2020

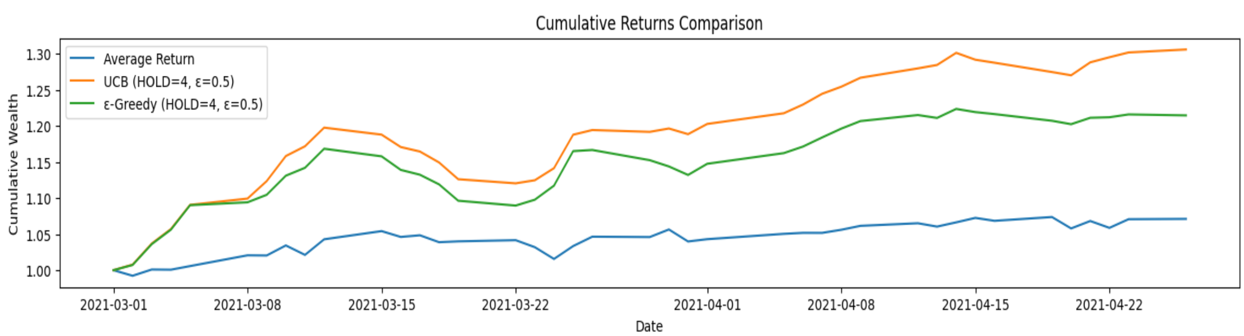

Figure 18: Comparing Cumulative Return of Average return, UCB and greedy algorithm during March & April 2021

In both the crisis period (March–April 2020) and the bull market period (March–April 2021), the UCB and ε-greedy algorithms outperform a naive average-return strategy because they explore and exploit for better actions. During the crisis, exploration tends to reduce the cumulative return, as randomly selecting assets increases the likelihood of exposure to sharply falling stocks. In this environment, a pure greedy approach (ε = 0) that sticks to historically best-performing assets preserves wealth more effectively, and shows that "sticking to what we know" is safer than exploring. Conversely, in the bull market, exploration and longer holding periods improve returns because historical returns are more predictive and trends can be exploited. UCB and ε-greedy benefit from both relying on historical performance and allowing for controlled risk-taking. The holding period (HOLD) determines how long an asset is kept before reallocation; shorter HOLD periods limit exposure in volatile markets, while longer HOLD periods allow gains to compound in trending markets. Overall, these results illustrate that the effectiveness of exploration and holding strategies is highly dependent on market conditions: exploitation is safer in volatile periods, whereas a combination of exploitation and controlled exploration is profitable in stable and upward-trending markets.

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